2007 F=ma Exam: Problem 25

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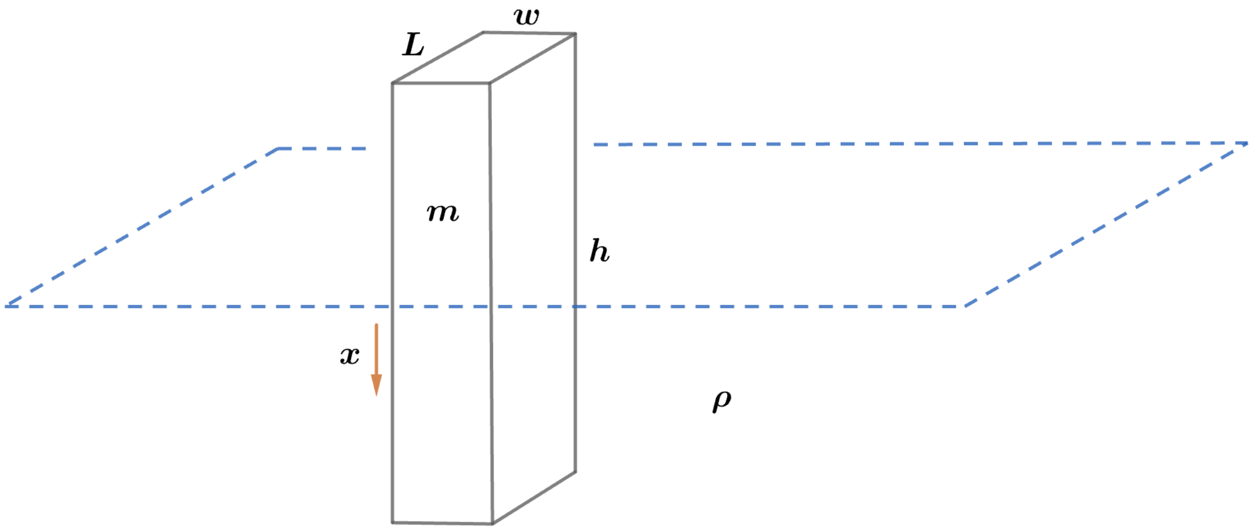

Starting at equilibrium, suppose we push the pogo down by a distance x. Then the buoyant force increases by the weight of the water displaced:

$$F_b = \rho V_{\text{dis}} g = \rho(Lwx)g$$

Applying Newton's 2nd law to the pogo (taking downwards to be positive), we have

$$F_{\text{net}} = -F_b = -\rho Lwgx = ma$$

By definition of acceleration, $a = \ddot{x}$ so

$$m\ddot{x} = -\rho Lwgx$$
$$\ddot{x} = -\frac{\rho Lwg}{m}x$$

This is of the form of simple harmonic motion $\ddot{z} = -\omega^2 z$ so we can identify the angular frequency ω as

$$\omega = \sqrt{\frac{\rho Lwg}{m}}$$

The period is given by

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{\rho Lwg}}$$

so the answer is $\boxed{\text{D}}$.